

AI Expert Algorithm — Final Exam (2026)

Topic: Human pose estimation

ANSWER KEY — INSTRUCTOR ONLY

ID: _____

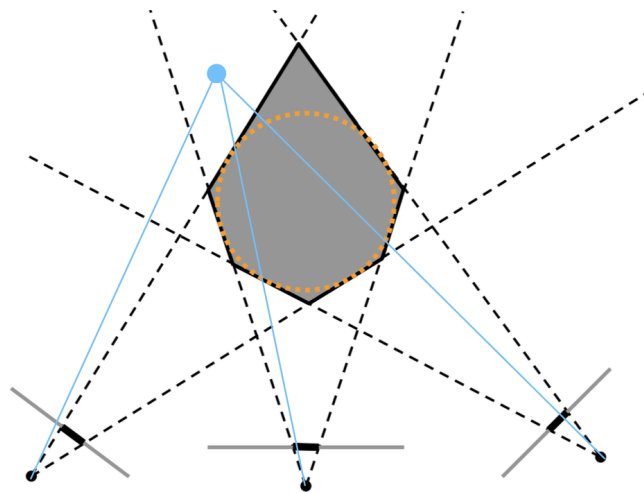
Name: _____

Instructions: Write your answers inside the boxes provided.

Question 1

[Difficulty: Medium]

Visual Hull (Shape from Silhouette). Given a 3D point X , along with several calibrated camera views and their input image silhouette masks (foreground vs. background), explain how to decide whether X is occupied or vacant. (Equations are not required.) (20 pts total)



Project the 3D point X into each camera image using that camera's known calibration (projection matrix), giving a 2D pixel location per view. Check each view's silhouette mask at that pixel: it falls either inside the foreground (silhouette) or in the background. X is declared occupied only if its projection lands *inside* the silhouette in *all* views.

Question 2:**[Difficulty: Hard]**

For the three 3D shape representations covered in the lecture, give a simple description and state one key advantage and one key disadvantage of each. (Point Cloud 6 pts, Voxel 6 pts, Mesh 6 pts; 18 pts total, 2pts for each item)

(a) Point Cloud

Description: an unordered set of 3D points $\{(x_i, y_i, z_i)\}$.

Advantage: easy to obtain directly from sensors (stereo/depth); represents arbitrary shapes freely.

Disadvantage: no neighbor/connectivity relations, so adjacency and surface/topology are hard to define; hard to compare two clouds.

(b) Voxel

Description: 3D space discretized into a regular grid of volume elements.

Advantage: well-defined neighbors; a 3D extension of the 2D image, suits 3D convolution.

Disadvantage: cubic memory $O(n^3)$; Manhattan-world (axis-aligned) bias.

(c) Mesh

Description: vertices plus faces (connectivity) defining a surface.

Advantage: represents the surface directly and is ideal for rendering (graphics standard).

Disadvantage: fixed topology makes reconstruction hard; self-intersection issues; higher resolution needs more vertices.

— End of Exam —